Coupled DHAK–PPK2 Assay with ATP Regeneration

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol tests whether a two-enzyme system regenerates its own ATP. DHAK phosphorylates dihydroxyacetone (DHA) to dihydroxyacetone phosphate (DHAP), consuming ATP and releasing ADP; PPK2 converts that ADP back to ATP using polyphosphate (polyP) as the phosphate donor. The assay has two separate reactions. The regeneration reaction runs for hours and is sampled over time. Each sample is then measured in a second, independent endpoint reaction, in which α-GDH and TIM convert the DHAP present into an equivalent amount of oxidised NADH, read as a fall in absorbance at 340 nm. The read-out is the DHAP formed divided by the ATP supplied at the start: a ratio above one shows that each adenine nucleotide was used more than once. Concentrations below are final concentrations in each reaction; scale the volumes to the format used.

Materials list

- DHAK, as cell-free extract or purified enzyme, of known activity (U/mL)
- PPK2, of known activity (U/mL)
- Dihydroxyacetone (DHA)
- Polyphosphate (polyP), of defined chain length
- ATP
- $MgCl_2$
- Reaction buffer, **50 mM**, at the working pH of both enzymes
- NADH, prepared fresh and kept on ice in the dark
- α-GDH and TIM coupling enzymes
- DHAP standard, for calibration
- Temperature-controlled shaker
- Spectrophotometer or plate reader at **340 nm**
- Freezer and ice, for stopping and storing samples

Steps · Reaction 1: regeneration reaction

1. Prepare the reaction with the following final concentrations:
    a. Buffer, **50 mM**, at the working pH of both enzymes
    b. DHA, **50 mM**
    c. ATP, **1 mM**
    d. $MgCl_2$, **10 mM**
    e. polyP, **75 mM** as phosphate monomer
    f. DHAK, **10 U/mL**
    g. PPK2, **20 U/mL**
2. The ATP sets the experiment. At **1 mM against 50 mM DHA** the nucleotide is limiting by a factor of 50, so any DHAP above 1 mM can only come from regeneration. Keep this ratio if the concentrations are changed.

3. The $MgCl_2$ is in excess over the ATP, since both enzymes act on the Mg-nucleotide complex. PolyP also chelates Mg^{2+}, so raise the $MgCl_2$ to **15–20 mM** if the polyP is increased.
4. The polyP is in excess over the DHAP that the DHA can yield, expressed as phosphate monomer, so that the phosphate donor does not become limiting before the DHA does.
5. The DHAK at **10 U/mL** converts the DHA over a few hours, slowly enough that the nucleotide pool must turn over many times rather than being consumed in a single burst.
6. The PPK2 is in excess over the DHAK by activity, so that ADP is recycled at least as fast as DHAK produces it. If PPK2 is limiting, the measurement reports PPK2 and not the regeneration capacity of the system. Confirm this by repeating the assay at **40 U/mL PPK2**: the DHAP formed should not change.
7. Prepare a **control reaction** that is identical but contains buffer in place of PPK2. Without regeneration, DHAP formation stops once the initial 1 mM ATP is spent, which separates the contribution of PPK2 from everything else in the mixture.
8. If polyP needs a pre-incubation with the enzyme or the Mg^{2+}, carry it out before starting the reaction and record the time.
9. Start the reaction by adding the remaining substrate and incubate with shaking at a controlled temperature, typically **25–30 °C**.
10. Take samples at several time points, for example **0, 1, 3, 5 and 7 h**. The zero-time sample fixes the baseline and should contain no DHAP.
11. **Freeze each sample immediately** to stop the reaction. All samples are measured together in Reaction 2.

Steps · Reaction 2: endpoint DHAP measurement (α-GDH/TIM)

1. This is a second, independent reaction. It does not follow the regeneration reaction in real time; it measures how much DHAP each frozen sample contains.
2. Prepare each well or cuvette with the following final concentrations:
    a. Buffer, **50 mM**
    b. NADH, **0.3 mM**
    c. Sample, diluted **1:200** from Reaction 1
    d. α-GDH, **1–2 U/mL**, added last
    e. TIM, **1–2 U/mL**, added last
3. Because α-GDH oxidises one NADH per DHAP, the **0.3 mM NADH is the detection ceiling** of the well. A sample above it saturates and cannot be quantified.
4. The **1:200 dilution** places that ceiling at **60 mM DHAP** in reaction terms, which covers the full 50 mM that the DHA can yield and sits far above the 1 mM ATP. Adjust the dilution if the concentrations in Reaction 1 are changed, and keep the ceiling well above the ATP, beyond any pipetting error, so that a saturated signal still proves that DHAP exceeded ATP.

5. Prepare DHAP standards at **0.05, 0.1, 0.2 and 0.3 mM** in the well, which spans the measurable range. DHAP is unstable in solution, so prepare them fresh, keep them on ice, and discard a standard that gives no signal.
6. Dispense the buffer and NADH base mix into one well or cuvette per sample, per standard and per blank.
7. Thaw the samples and add the same volume of each sample, standard or blank to its well.
8. Read **A_{340} before** adding the coupling enzymes. With 0.3 mM NADH this is about **1.9** at a 1 cm path length, proportionally lower in a microplate well, and should be similar across wells.
9. Add the **α-GDH and TIM** to every well and mix.
10. Read **A_{340} after** the signal has stopped falling, so that all the DHAP present has been converted.
11. Calculate **ΔA_{340} = A_{340} before − A_{340} after** for each well and subtract the blank.
12. Convert ΔA_{340} into DHAP. Against the standard curve, read the concentration directly. Without standards, use **ε_{340} = 6220 M^{-1} cm^{-1}** for NADH and the real path length of the vessel, which is not 1 cm in a microplate well. Multiply by the dilution factor, **200**, to obtain the DHAP concentration in Reaction 1.
13. Plot DHAP against time for the complete reaction and for the control without PPK2.
14. Divide the DHAP formed by the ATP supplied at the start. A ratio **above 1** shows turnover of the adenine nucleotide pool and is the proof of concept for the coupled system.

Troubleshooting

- **Samples saturate the assay:** Any sample whose ΔA_{340} equals complete NADH oxidation gives only a lower bound for DHAP. Dilute it further and measure again. Saturation at every time point still demonstrates regeneration if the ceiling is above the ATP concentration, but the time course cannot be resolved.
- **Detection ceiling too close to the ATP concentration:** A margin of a few per cent falls within pipetting error and does not support the conclusion. Raise the NADH concentration or dilute the samples further.
- **Signal too small to quantify:** If ΔA_{340} falls below about 0.05, the sample is too dilute for the ceiling chosen. Use a lower dilution and keep the expected DHAP between roughly a fifth and two thirds of the NADH in the well.
- **DHAP at time zero:** Check for carry-over, or for DHAP already present in the enzyme preparation.
- **No DHAP in the control without PPK2:** The control should form DHAP until the initial ATP is spent. If it forms none, DHAK was inactive and the complete reaction cannot be interpreted either.

- **Control and complete reaction indistinguishable:** PPK2 is inactive, the polyP is not a usable donor for it, or the ATP was not limiting. Verify that the ATP is substoichiometric with respect to the DHAP that can be formed.
- **Reaction stalls early:** PolyP sequesters Mg^{2+}, which both enzymes need as the Mg-nucleotide complex. Raise the $MgCl_2$ and check that the pH has not drifted, since polyP hydrolysis releases phosphate.
- **Baseline falls before α-GDH is added:** The sample contains NADH-oxidising enzymes, which is more likely with cell-free extract. Record this slope and subtract it, or dilute the sample further.
- **Unstable NADH:** NADH degrades in solution. Prepare it fresh, keep it cold and in the dark, and confirm that the initial A_{340} matches the expected concentration.
- **Records needed for reproducibility:** Report the activity of DHAK and PPK2 in U/mL, the polyP chain length and concentration as phosphate monomer, the buffer and pH, the temperature and agitation, the sampling times and the dilution factor of the endpoint assay.